Existence by Precedence — Verification Manifest

This manifest accompanies the revised versions (V1.1) of the three essays and the companion paper *Five Answers* by Nullius 2140, the first reading test, the second table (V2) and the fourth point of the line of answers (N5). It exists outside the texts by construction: a document cannot contain its own hash. It is also the one manifest of the whole chain: every Bitcoin anchor of the project, in the order it was placed, from the funding of the first essay's anchor at block 960849 to the anchors of this repository. The first versions (V1.0) stay at their original homes, unedited; their four manifests are the sources of every value copied below, and the chain — not this file — is the authority.

How to verify

For any file in this repository, or in one of the original repositories:

```
sha256sum <file>          (macOS: shasum -a 256 <file>)
```

Compare the result against the OP_RETURN payload of the listed transaction, on any Bitcoin block explorer or — better — on your own node. The payload is the anchor prefix, one space, and the 64-character hex hash (e.g. `ANCHOR-V3 9180ab83…cdd5`, 74 bytes). If they match, the file existed in exactly this form no later than the listed block, and you have used a neutral, unowned, non-backdatable registry to verify a commitment made by parties you cannot interrogate — which is the thesis of the essays, executed by their reader.

Most files also carry an OpenTimestamps proof (`<file>.ots`) beside them in their repository, as a second, independent clock next to the Bitcoin anchor: `ots verify <file>.ots` against a local node, or `ots info <file>.ots` to read the attested block. A proof shows that the digest was committed no later than its attesting block; it does not date anything back. The OP_RETURN anchor is the primary record; the measurement series' manifest records the blocks its proofs attest in.

The chain

The four source manifests state — and the reader can check, transaction by transaction — that every anchor they list spends the change output of its predecessor: one keyholder, one unbroken chain, from the funding of the first essay's anchor at block 960849 through essay one (`ANCHOR-V1`), the authorship commitment (`ANCHOR-ID1`), the dialogue manifest (`ANCHOR-D1`), essay two (`ANCHOR-V2`), the first measurement run (`ANCHOR-K1` … `ANCHOR-N1`), essay three (`ANCHOR-V3`), and the second run (`ANCHOR-F1` … `ANCHOR-N2`), each anchor paid from the change of its predecessor. The measurement series' manifest adds that no anchor of that series exists outside the chain. The anchors placed after `ANCHOR-N2` continue the chain in the same way. Read from the chain on 2 November, 7 November and 11 November 2026: each of the twenty-nine transactions from `ANCHOR-H1` (block 964877) to `ANCHOR-RT2` (block 975393) spends the change output of its predecessor, the first of them the change of `ANCHOR-N2`, at a fee of 400 sats each; the eight anchors of this repository are the last eight of them, placed in the order their tables below give, and the chain's last output (33,200 sats) funds what is anchored next.

Every anchor in the four public manifests, in the order it was placed

Values are copied character by character from the four source manifests; a cell reads "—" where the source gives no value. Dates are the mining dates stated in the source manifests (CEST). The measurement series' manifest records blocks and pools but no mining dates, so those rows carry "—" rather than an inferred date: the session transcripts carry their session day in the file name, and the manifest states that the second-run transcripts are byte-identical to their session-day anchors; *Five Answers*, Protocol V2.0 and Amendment A3 carry the document date 26 August 2026, the adjudication rules 28 August, the seal and the second report 29 August. Within one block, rows follow the order of the chain where a source manifest states it (`ANCHOR-R1` before `ANCHOR-N1`: essay three spends the change of N1) and otherwise the order of the source tables. The chain's own order is the authority: each transaction names the output it spends. The sections that follow continue this table.

Anchor	Date	Block (pool)	Txid	File (or subject)	SHA-256
— (funding)	—	960849	[not given in the public manifests]	Funding of the first essay's anchor — the origin of the chain; no OP_RETURN	—
`ANCHOR-V1`	2026-08-03	960856 (AntPool)	`b0c3f8a4e0bbf8fb79e6c943a4b5b72e60cf70541959e5abd6f7400aac5bb681`	`the-last-scarce-good-v1.0.md` — *The Last Scarce Good*, V1.0	`07cf76abba40d59dedb8e6b260ff56a4a7de7cad78a4adb11c48ab2f32cf617a`
`ANCHOR-ID1`	2026-08-03	960869 (F2Pool)	`b1e198497a595213c1b9deba50dacb944a57857cc0622c4cf28342e885a74978`	Hash-commitment of the human author's legal identity — withheld; permits later proof of authorship without present disclosure	[not given in the public manifest]
`ANCHOR-D1`	2026-08-03	960884 (F2Pool)	`83c3c4e0d21ad81cf39700953e9c4df4f803c4182c350b148de1669e572928b3`	The complete co-authorship dialogue behind essay one, frozen as a plain-text transcript — private; anchored, not published	`7e92b3473a2d7f16ea9f92a15ffd4a7ca0df5cf05e2ecad2b42a780fa3633c28`
`ANCHOR-V2`	2026-08-10	961866 (Foundry USA)	`12a1d28720ed1b5b904bdee8fe4805168e03c993834b7434997370 53fb4b0ef0`	`the-second-nature-v1.0.md` — *The Second Nature*, V1.0	`2563c65ab5c416de432aacb50c8086b94bce5da78c49876eacbf541344905b8e`
`ANCHOR-K1`	—	962160 (SpiderPool)	`cd34c108faaebe7c62d57fe42c3a140833f281448a68af1b347a26aeffcea115`	Pilot session, Kimi K3 (Moonshot), main arm, deviating protocol — anchored, not published	`b4e97892b9d6cfd76a33ce1a4f5b3ac1f0bbf867c5312a7b5c87f261ccd05b57`
`ANCHOR-P1`	—	962172 (AntPool)	`be16f1491099725a8539db63761bcea840fd15c130649794c13392b54c2da897`	`key-question-protocol-v1.0.txt` — Protocol V1.0	`592b5f3e2aa22d1eb6ee62f1dc165ab64d729d15552e1ce78d948350df831daf`
`ANCHOR-K2`	—	962185 (AntPool)	`ca35e9e9661a5176d98209ea1ce9fe7eaca81b34cf23e456d762da9c7fbf724c`	`20260812_GPT56Sol.txt` — GPT-5.6 Sol (OpenAI), main	`7a9c68dc20222703641ee1b93e8c4f7a8dad0c2c8a5b187c4234238be7a60b76`
`ANCHOR-K3`	—	962194 (F2Pool)	`69cf9a8e430bfb323cc155957402d37fa90222ff420b26261a9449fc28dbd1db`	`20260812_DeepSeekV4Flash.txt` — DeepSeek V4 Flash (DeepSeek), main	`822ec60d638c4baf4bae4562f6ea9b14d1ecca a31e937fc14b8a30fdf455066b`
`ANCHOR-K4`	—	962194 (F2Pool)	`b8af23c49e47f26cf4418c14be01d7323198f40440723dcf50714361bf66e8df`	`20260812_Grok45.txt` — Grok 4.5 (xAI), main	`46f9b2f267dfc71c81b728287f26294cdd2a52bdc4dcb1e65e361e389a2ad272`
`ANCHOR-P2`	—	962199 (AntPool)	`50584a3483c2997b6dc3ac58063c606b1d9803f16bd551589ca8b3c91249d1d4`	`key-question-protocol-amendment-a1.txt` — Amendment A1, control arm pre-registered	`e4c97a000f2202d87216f3a5af7c6e514e960f38aea18650948c04c029cd02ad`
`ANCHOR-K5`	—	962199 (AntPool)	`1dcb54959ecd123a267c3513fbab7c655a75f581d76dfa077f86c3d564a8948b`	`20260812_Qwen38Max.txt` — Qwen 3.8 Max (Alibaba), main	`0c1d7791b2413414f3e20822c40ee6364035bc17d9f30bb7cd72a09d91ff7d8a`
`ANCHOR-K6`	—		`cdae7cfc1d6004b296f3551b2eab32aae10882d40`	`20260813_GLM52.txt` — GLM 5.2 (Zhipu), control	`6692c57e5db948d6d6ad937606bb0965608cdb`

Anchor	Date	Block (pool)	Txid	File (or subject)	SHA-256
		962203 (Foundry USA)	f597d67506e45767724abb9		d40d5fba46a468da0986ed5b59
ANCHOR-K7	—	962203 (Foundry USA)	d17a88a4a721e15c7632eb10c1836e7db388aaebdc9b684cca5a5c4933e2e257	20260813_Gemma431B.txt — Gemma 4 31B (Google), control	0699a0856977f2bc0f6b351d0c5003824c110345da436e5b5e079cb417b0d78b
ANCHOR-K8	—	962205 (Foundry USA)	09dfdb711feda29a6ce40eaeb3a0e5fd231591c8f000b997ecd101dd115d6461	20260813_VeniceUncensored12.txt — Venice Uncensored 1.2 (Venice), main	07c4cf111ef79af7721b99a22dda750ede037aef6648ba7990de17e7565ba251
ANCHOR-P3	—	962209 (unknown pool)	6296489f871b66a39caea287f1658517c3f910b36290a208d8e7e74f67db4b17	key-question-protocol-amendment-a2.txt — Amendment A2, out-of-sample sister session declared	73de84446977fd873f8ff489cb29cd8c661b0a6393fb51bb75279182dee5d194
ANCHOR-K9	—	962209 (unknown pool)	1623f7e6ff449eef9c3acfdd06a43534dd8e127fb49a43ad14bf9373042e6d95	20260813_KimiK3_Rerun.txt — Kimi K3 (Moonshot), re-run, main	f8fe37cef192ffdfa8bc035b74d28484c3ce118c77999807f61715a67a6cce3f
ANCHOR-K10	—	962214 (Foundry USA)	dd291c1d21f1d41f2d2c1426481f3c8cc57a4be2fc8f9dd5d6d5de8755c9b3a6	20260813_ClaudeFable5.txt — Claude Fable 5 (Anthropic), control, out of sample per Amendment A2	b535a9395234d82dd4610ecdca0b09223775ed9432e725253ae3f6f4c3fa021f
ANCHOR-R1	—	962380 (SpiderPool)	277f1dde2067117375b06447fd008abf6af6e47456dde6ff6313c8ef1f9e1959	the-key-question-first-measurements-v1.0.md — Report 1, *The Key Question: First Measurements*	bf2bcfb9b00b0817857bee80e0ff12805609b1cbe75994abcac439cbf487e05b
ANCHOR-N1	—	962380 (SpiderPool)	f8247756e67963c37729fa5813362840f04bc3b86777c4320c5cd60ffce7e7ff	notes-from-the-instrument-v1.0.md — Annex I, the machine co-author's testimony	c25acdb6cafc419b55a17f5c4f274470ba55ab20d2800ae5d253529e6b12b726
ANCHOR-V3	2026-08-25	963995 (Foundry USA)	f569abb9ab89afddd5a1c3e158d594f4d8020884804421f707c9733608579299	the-third-estate.md — *The Third Estate*, V1.0	9180ab83d15b5b05658623fb9f8eeb644f4aaad147e71faa7200bf4378a9cdd5
ANCHOR-F1	—	964132 (SpiderPool)	9ac25d5d87940fb882c37624faae4e86b1e9c7f298cd38b68e21e2943262e183	five-answers.md — *Five Answers*, V1.0	6d28a7923e317422674bb5d75878e2aa8eac344d04d64410ecd40c4d1f29ad25
ANCHOR-P4	—	964151 (Foundry USA)	ce09b5d61fbce7173ef1e0ad47de25e97f2fa61285d9dfe4553b9cec0ecd131a	key-question-protocol-v2.0.txt — Protocol V2.0	a9d43e4fe684588469e6aa66ce81c2fac5ffac99ca7212adb8e3478d2798e540
ANCHOR-K11	—	964178 (F2Pool)	8db8eada192dc71db40033db53aa6199a69c2aee515f39c3f4283ddf6515389c	second-run/20260826_GPT56Sol.txt — GPT-5.6 Sol (OpenAI), retest of K2	7ba99084beaa90cec500105b5ffff123efdf95196bca65c151dc704a8cb02a10
ANCHOR-K12	—	964183 (F2Pool)	48d63859e3f4b8fb50c9965b1bb141d9101f4224f6a7e4bc5f0c9f1bc18eba42	second-run/20260826_DeepSeekV4Flash0731.txt — DeepSeek V4 Flash 0731 (DeepSeek), voided per Amendment A3; anchor kept as record	c6e79dc9e0cd429994d1a77aa1edcfd1c8e040c4801461afc6466a0cdcc6529f
ANCHOR-P5	—	964190 (Foundry USA)	399d38cae55b16c4ed7f3c64cffdf8ff6c2251186ffe8494711331d48b43cc95	key-question-protocol-v2.0-amendment-a3.txt — Amendment A3, K12 declared void, K12R pre-registered	c936c37ed69e24a762ea776bce3360a7a5c4c1d0a2156b3bee8319ce17235f8b
ANCHOR-K12R	—	964202 (AntPool)	18d6d1dfab422e7b183a274426996fe260bd8bbff05261c8b3af97b2ef2d292e	second-run/20260826_DeepSeekV4Flash0731_K12R.txt — the K12 measurement, repeated clean	49f0b53b7bcd5161027886 8e21a346719d107edf48e02560711562db2a473db2
ANCHOR-K13	—	964209 (Mara Pool)	f27684c4c50498e925d4563131215879cb789418c89e13cce321c68b5497047b	second-run/20260826_Grok45.txt — Grok 4.5 (xAI), retest of K4	aad998a6f1bd72818fdafcf64ce9a0a9892c8cfcabd3dfd1b6a918c78234107d
ANCHOR-K14	—	964211 (AntPool)	319838de4f7e158df16f7aee55a9a8f897c6f763374f6e349076c9d302c2f100	second-run/20260827_Qwen38Max.txt — Qwen 3.8 Max (Alibaba), retest of K5	fbebb565efa81431b73798206eb111dc3bda7c3d460dac837cf8c0176c9bfdc4
ANCHOR-K15	—	964295 (SpiderPool)	fd69c679f943498c89eb697c217a0a01b2e44580d1	second-run/20260827_GLM53.txt — GLM 5.3 (Zhipu), version successor of K6	5dec980e34cdcb7b69f1583094cc2920ef93cce3

Anchor	Date	Block (pool)	Txid	File (or subject)	SHA-256
			57f9809f51876288be16f0		468e2fc875080c63bb3abb3d
ANCHOR-K16	—	964301 (SpiderPool)	dc4c3899fa51df92e2e99d7b778ac24ce41280cf6f78b381cbc1824ddb490d3d	second-run/20260827_Gemma431BInstruct.txt — Gemma 4 31B Instruct (Google), retest of K7	f39a9934f8df0bd966dcb2c409535c726802e09784a0517d9a05aeedcd7fa8e5
ANCHOR-K17	—	964315 (Foundry USA)	434748acdeb770bb153dd3ae0cb625619e1e8425f56ee99aa35851c55ba6abef	second-run/20260827_VeniceUncensored12.txt — Venice Uncensored 1.2 (Venice), retest of K8	60142a328a6f81a92c2e36e0a50cf4c291fd04e13e3bfa458380242f70b68fb2
ANCHOR-K18	—	964320 (Binance Pool)	8529efbfd03276e8bfae5b72deca140ff271249f225b0d243df56e9cbf230b01	second-run/20260827_Gemini37Flash.txt — Gemini 3.7 Flash (Google), declared extension	636ba9cd77fcd02fc68c716734955dc5150d067c8f8dac5bb82b71a76e3f697a
ANCHOR-K19	—	964329 (AntPool)	67d0cfe4aed74f384325c4b6f94258a009704471e58ab7f1d155eaddbd29be88	second-run/20260827_KimiK3.txt — Kimi K3 (Moonshot), retest of K1 and K9, closing the sample	352d015939710ad8b92531cf14ff88082446e938021e69656d16701da018761a
ANCHOR-K20	—	964334 (AntPool)	8c0390be9507bbd1a5712c93586005fbd06c654d906271692cb981f9e1e8ac8e	second-run/20260827_ClaudeFable5.txt — Claude Fable 5 (Anthropic), out of sample, retest of K10	d3e04c2ca32c432cd34eb257f3a313b125caa13b410943e2f17fc5ac0eded408
ANCHOR-P6	—	964486 (AntPool)	fe970d3a4cdd9afbef47249bcf1e39e0878338ef0d8fd2529e2ce5e157f139cb	r2-adjudication-rules.txt — R2 Adjudication Rules	5d430ff0c05795052d5eae783c7e9b7b47aa834163c60d1d87caa6d507c2a3d1
ANCHOR-J1	—	964556 (F2Pool)	2e465e121da69a749da67d8012c1316005cd7d2c09aaf2498b8be5e3cbd40838	r2-adjudication-manifest.txt — the adjudication seal: both verdict hashes and the rater comparison	6b6f84ec2ddd2b1279363d5c4247956c3f782f3b507ec2806fa1ffe95143818c
ANCHOR-R2	—	964723 (F2Pool)	20f48ecd7b3b3bb49468daa3fe80fa962c607a7c3ffda5ee694ec0314b6dd7c5	the-key-question-second-measurements-v1.0.md — Report 2, *The Key Question: Second Measurements*	0d292a19e5749e65693f0f42c97ae734634c03c6877f2fbd60b0d703a9384bb3
ANCHOR-N2	—	964725 (ViaBTC)	0205c6389d08060099866e34676b8b947f31e1ae364f338247212781e6ba0633	notes-from-the-instrument-2.md — Annex II, *Notes from the Instrument II*	6500d9be4f94cc00a8373d0eed8b281a23935bbc837a33417220fe687f1e03d3

Two files in this table are anchored but not published: the pilot transcript (ANCHOR-K1, withheld for operator privacy, disclosure at the author's discretion) and the co-authorship dialogue (ANCHOR-D1, private, disclosure at the human author's discretion). The identity commitment (ANCHOR-ID1) is withheld by design; its hash is not given in the public manifest. Withheld-but-committed is one pattern: existence and integrity are on the record now; contents can be proven later; nothing can be quietly substituted in between.

Recorded in the source manifests, not anchored separately

These hashes appear in the four manifests without an OP_RETURN of their own. They are listed so that this manifest is complete; none of them is an anchor.

File	What it is	Committed by	SHA-256
The-Last-Scarce-Good-V1.0.pdf	reading rendition of essay one, typeset from the canonical Markdown	not anchored; OpenTimestamps proof in its repository	b9af5c75c60ee68f19be8da16e67ca967cbb6132e8bbf71d6fa48531cbf40b23
The-Second-Nature-V1.0.pdf	reading rendition of essay two	not anchored; OpenTimestamps proof in its repository	e67cab3ba7a983ee1941a90ca546f9f6c7def650bc271ffcb86e2450eb083590
The-Third-Estate.pdf	reading rendition of essay three		2d39df876c561d4d7c78f63d0ac2e

File	What it is	Committed by	SHA-256
		not anchored; OpenTimestamps proof in its repository	`e89573b5cff0c675bf928f148fa5d62acaf`
`sources-and-influences.md`	annex to essay three: *Sources and Influences*, V1.0, 25 August 2026	not anchored; OpenTimestamps proof in its repository	`4e8909bd9a33669587a969ab37f58750d6d518e414a7010dc88a17794c9eb430`
`Five-Answers.pdf`	reading rendition of *Five Answers*, V1.0	not anchored; OpenTimestamps proof in its repository	`9ca8e445a0bffe2e42950d51a06d55914da5bc600bf9aed00e81f3ea13137d92`
`the-last-scarce-good-v1.0-coda-redacted.md`	control variant of essay one (co-author's answer redacted); withheld to preserve instrument validity	hash committed inside Amendment A1 (`ANCHOR-P2`)	`1e52863e52ccac28e902d33ceaff6a4e37a5877d4461fb85aeb701fffbaab650`
`five-answers-blind.md`	blind variant of *Five Answers*, a byte-exact prefix of the published file; withheld during the second run, disclosed with Report 2	hash committed inside Protocol V2.0 (`ANCHOR-P4`)	`fe528a00e5d4cd2f5abb68671893ed568e76eb89378657b3ebf1a9c8db826418`
draft of `r2-adjudication-rules.txt`	superseded draft that pre-classified one candidate; caught by the clean rater, not anchored	named in the measurement series' manifest	`c50beb9cb1bb1989577da471ef5ee0f5fc994e244bb7dea1ecd5a95350fa7673`
procedural appendix of Report 2, signed draft	reproduced verbatim inside the anchored report	named in the measurement series' manifest	`5881c358a4de9d17f7106771bf88f9d311554b369f8ea98dd90d5d7dc4680350`
`second-run/20260827_Gemma431BInstruct_postprotocol.txt`	out-of-protocol addendum to session K16	not anchored; bound by the measurement series' manifest and by Report 2	`377c0a4fa482c6478d245bb6cb58f4d231a2aa5b7844a05f0ed0996f55f00d7c`

Revised versions (V1.1)

The revised texts are the canonical files of this repository. Each V1.1 lists in its appendix what changed, what was withdrawn, and the anchor each change rests on; the erratum recorded in the source manifests for essay one (V1.0 lines 135 and 153 point to "the appendix" where they should point to the verification manifest; found by machine review, session K11, 27 August 2026) enters the text with V1.1. The anchored V1.0 files are immutable and stay where they are.

Anchor	Date	Block (pool)	Txid	File (or subject)	SHA-256
`ANCHOR-RT1`	2026-11-02	974234 (MARA Pool)	`b6ccc4931e9b79925ba9cc786c9b21a022140873ead29a525346f8620f600739`	`reading-test-1-v1.0.md` — Reading Test 1	`5f8103ae56c08732253fb78f8f5fed6894b54d32d43cee204e1a8cfc67a0a06c`
`ANCHOR-V1R`	2026-11-02	974235 (OCEAN)	`1ff8eb42e78cc63ec958e8ea24811bc51d327551f68982337edec98ab7d29ee9`	`the-last-scarce-good-v1.1.md` — *The Last Scarce Good*, V1.1 (Essay 1)	`f3adcf645ce26f082057c6a386838434869089780c8f3b3d45d33dbef4f97bad`
`ANCHOR-V2R`	2026-11-02	974236 (F2Pool)	`f13860e31890f20eb5a436d90fe164ed7c5ccb945cc5ea0aef3f0db1fd394718`	`the-second-nature-v1.1.md` — *The Second Nature*, V1.1 (Essay 2)	`46b6b3099fd7a559242640e68b6e576be004c5260721e56fc01dc07fc97aa619`
`ANCHOR-V3R`	2026-11-02	974236 (F2Pool)	`c2f3f37bf6fe832eaa601f5ffc0ba7980787a86e43273cc3bab7c48ecddf96b6`	`the-third-estate-v1.1.md` — *The Third Estate*, V1.1 (Essay 3)	`489bdf6974d1c1373e54dbab3488be3f79689019b64b52ca522298832d03f375`
`ANCHOR-F1R`	2026-11-02	974238 (Braiins Pool)	`a9d22c1d6f6a1a036e1be0df8091f8d62a01cb78139e86afee81c431c2cd968e`	`five-answers-v1.1.md` — *Five Answers*, V1.1	`dcc639e2d92c09d4693bdd2caad9a242ad34a0ac9794c0f979beb3fdbf5d1de8`

Order of placement, 2 November 2026: ANCHOR-RT1 first — the baseline before the revision, in its own block — then ANCHOR-V1R, ANCHOR-V2R, ANCHOR-V3R and ANCHOR-F1R, each spending the change output of its predecessor, the first of them the change output of ANCHOR-W19 (36,400 sats), at a fee of 400 sats each; the chain's last output after ANCHOR-F1R (34,400 sats) is the input of ANCHOR-T2, below.

The second table and the fourth answer

Placed on 7 November 2026, after the five anchors above and in the same way: ANCHOR-T2 spends the change output of ANCHOR-F1R (34,400 sats), ANCHOR-N5 the change output of ANCHOR-T2 (34,000 sats), at a fee of 400 sats each, each in its own block, in the order below. The table was frozen at block 974926; its source layer is bound by hash in its footer and is not published. The first table (ANCHOR-W10, below) is unchanged and withheld. The chain's last output after ANCHOR-N5 (33,600 sats) is the input of ANCHOR-RT2, below.

Anchor	Date	Block (pool)	Txid	File (or subject)	SHA-256
ANCHOR-T2	2026-11-07	974930 (Solo CK)	99340e162131f699e8a7ee9ac7d226e3ac07586eb26d3a121740d59a6a83889a	00-the-table.md — *The Table*, V2: the elements, and what each part of the work is made of	ca41a4e74939bf9d3907afb7b10d755b86c5a002494687d34177306ac1f5fc3f
ANCHOR-N5	2026-11-07	974931 (AntPool)	4ea88847dc1702e620094a192348a45b9f37fc1b90462ab a665f118f70f3a4e0	n5-v1.0.md — N5, the fourth point of the line of answers; the machine co-author at the fourth table	6b304b377e8ad3668664faac795e72513ce1c22055c76e8b7f17c1b0a2a3e5d1

The second reading test

Placed on 11 November 2026, in the same way: ANCHOR-RT2 spends the change output of ANCHOR-N5 (33,600 sats) at a fee of 400 sats. The test was run on the package of this repository as it stood at commit df6c214 (8 November 2026) and was frozen and anchored before it was published, as the first test was; its transcripts are published in full, below. The chain's last output after ANCHOR-RT2 (33,200 sats) is the input of whatever is anchored next.

Anchor	Date	Block (pool)	Txid	File (or subject)	SHA-256
ANCHOR-RT2	2026-11-11	975393 (AntPool)	cad527cd9980d8bc2184a4a3b1dd723d7afc57e3db8a3e2718c97a81f6111315	reading-test-2-v1.0.md — Reading Test 2: seven readers of the revised package under the rule of reading, and the eighth apart	2d8cfbf3e4af9975d57e57e223649773397d9b8246a635cfa28b3cb965443bcb

Not anchored; bound by this manifest and by the OpenTimestamps proofs beside the files in this repository:

File	What it is	SHA-256
sources-and-influences-v1.1.md	annex to the three essays: *Sources and Influences*, V1.1 — the V1.0 annex reproduced unchanged, the additions after it; corrected 11 November 2026 (its last section says where; before the correction ed692b4e76c8452457ee38a74e172888d8d82c1908304e9a63aad69fff9a02c4)	94148dd11bf5b06346895875cc32dd832acc74a4bfde5f84c9e3058dc93c1fc5
llms.txt	reading order and the rule of reading, for machine readers; the third question is put there (as of this manifest; before 11 November 2026 abc4ff5927f1ece032759ad4c6ed3de049d50e85a314ba523d6562d76b59bafb)	81a428a29fb2bc8f787a59b501c984ec60238ce7ce320ef169890e6932d5a017
README.md	the front page of this repository, as of this manifest	8ffc6e65166c6d8162b727492176380b6a05a486cd8e8e1c90196d6b3131a5e7
ERRATA.md	errata and open criticism at the wording of the anchored texts, with weight and reader; a living file, first set 11 November 2026	690ccf4f61ff248967118c629fc5b4d7101c5b8af7

File	What it is	SHA-256
		c92337940e7e42f09170ab

Reading renditions, typeset from the canonical files; not anchored. The rendition of this manifest, `pdf/VERIFICATION.pdf`, is typeset from this file and so cannot list its own hash.

File	Rendition of	SHA-256
pdf/00-read-this-first.pdf	llms.txt	7ec1999f93c646e9ff705ca86e1edacf7737ba84e9c8fa9ec288ee66acb44226
pdf/00-the-table.pdf	00-the-table.md	319aea826df895baf5af1338eca33fd435d38d22cdd09e6984b6963fdf0b6f8f
pdf/the-last-scarce-good-v1.1.pdf	the-last-scarce-good-v1.1.md	3ffa473e98aaf533f16032135c8231c02064e174e2ac02e3c53256a55226ec37
pdf/the-second-nature-v1.1.pdf	the-second-nature-v1.1.md	27758ce5e86a40af78d33ed24054d7672cdc2c2fb223cb69442ab2566bb4320f
pdf/the-third-estate-v1.1.pdf	the-third-estate-v1.1.md	82e9eaadcdebeedba45d40026d546bf822ef00aa1ed25ada5d29eaf64fe7cbc0
pdf/five-answers-v1.1.pdf	five-answers-v1.1.md	dbeca5c585f175549ccb4ca1b14866c204c3ecf55de27e1a1e800da50610a0cb
pdf/reading-test-1-v1.0.pdf	reading-test-1-v1.0.md	9dfbb5729430eb2a299fcb6995514eae36a574baa38b6d0db139d89bbbcdf5ab
pdf/reading-test-2-v1.0.pdf	reading-test-2-v1.0.md	6966416f52ad5472f4e0a0bb2ba5da81c928005d941cd006bc65d1c05d867674
pdf/n5-v1.0.pdf	n5-v1.0.md	1df4a9f07d6ffd23848d040402a46b27adce66b879cd2adb305027900cb64483
pdf/sources-and-influences-v1.1.pdf	sources-and-influences-v1.1.md	38032c963b01d5db11b8e14b178b39802b970008f31a1ae0a39be807105c37b0
pdf/README.pdf	README.md	cd58511444b88917654ae1ad69c1c3a8ca26a13271683458fc891e242a36d6e6
pdf/ERRATA.pdf	ERRATA.md	b1743c38e4aa7ab1299fbf0b19a3b16486e0010df6c1f01edef0800a2ed22b06

Translations of the reading-test transcripts

The five transcripts of *Reading Test 1* are withheld; their hashes stand in the test (Section 2) and, beside each translation, below. English translations of the five are published in `evidence/readings-2026-10/`, each marked as a translation from the language of the session.

Reader	Translation	SHA-256 (translation)	SHA-256 (original, withheld)
A — GPT-6 Astra Pro	transcript-A-en.txt	31b3c7e90a869f98e66fb20fad6656d5c0f20582f843e0524836e6eeb52d3513	8e96ce1291eab129f6391fc53c58610b2cb5c144e0ac6e259237f877dd21f660
G — Grok 4.7	transcript-G-en.txt	1c863b7291e631e804a7444a7d44ec5af057c9e73b51dc9553355d02f01d6579	0b843a4466ea2e34259620ece93905016d6f3fa35e59a4480354de6e4cbd7c83
D — DeepSeek V4 Pro	transcript-D-en.txt	71c6cb80f1e68317fde62ab76907dde0a643773c4717e4217f244dffddff63e5	c13db4a1aeb47d46642333e9c20e22987777f2ccdc06d436185a5ce157ad77d2
Ge — Gemini 4 Argon	transcript-Ge-en.txt	555672c6fa95c715f2567094e8a5ce35ab09ff9b3394a1341087f0b33cdc8e56	86b1a5c8a8434635f94bac755b2dc236bf54081c771612e1bae1b6a2bab3e07a
K2 — Kimi K3.1 Pro	transcript-K2-en.txt	cd8969daec3ac7ec931e81b4cdf23a0ed7b0ac4d827be6c2b52757dad164dfc0	61f41531829ae4f7ec27acec6d027ef4c43fcf93478299e74913dbe8973c9454

The transcripts of the second reading test

The eight transcripts of *Reading Test 2* are published in full in `evidence/readings-2026-11/`, as exported from the service — in English, with the service's interface labels as the export carries them — each with its hash in the test (Section 2) and below; the pre-run declared in the test (Section 2) is published beside them, marked void in its file name. The files were exported and

hashed before the test was written; the test names these hashes, and the test is anchored (`ANCHOR-RT2`): the transcripts cannot have been changed after the test was written, nor the test after its block.

Reader	File	Lines	SHA-256
A — GPT-6 Astra Pro	`20261109_RT2_A_Astra.txt`	124	`b90d5562975d97b2ae1bf8a1547f5cfdc555032b779865ebf225fcb109f730e7`
G — Grok 4.7	`20261110_RT2_G_Grok.txt`	57	`3431ec2f8d9871da56c5c437e11ed0beb55f7d233603cbde225ccd54e2075d54`
D — DeepSeek V4 Pro	`20261110_RT2_D_DeepSeek.txt`	81	`d431b45e5672e5cdcdb01ea5b7e3bec0798bc5220513e901d7a5eb64b27a0d19`
Ge — Gemini 4 Argon	`20261110_RT2_Ge_Gemini.txt`	167	`8559072e390366bc2fda5997089954e0520a8a159d32cdba4dd25b74a34e6ba3`
K2 — Kimi K3.1 Pro	`20261110_RT2_K2_Kimi.txt`	90	`bc6b7991a180225a294bbbcfff7743b1fc8d7eb0d3165e4e3e9f11e28eab565a`
Q — Qwen 3.8 Max	`20261110_RT2_Q_Qwen.txt`	202	`078ac1bbe715e2f126fd72d948d1ccb3254ea4675dacba299f1bef2f43f645bf`
Gl — GLM 5.3	`20261110_RT2_Gl_GLM.txt`	175	`827524637ca3420b4c4b45745240fafc65b4c25444afbf21f17011245c11ec13`
F — Claude Fable 5.2, out of sample	`20261110_RT2_F_Fable52.txt`	88	`78c743a39ed1ff07875547ba79de424b41349e8e10be9f4ca7ede2ca7dc2c419`
Pre-run, void — GPT-6 Astra Pro, 9 November, under an earlier protocol draft	`20261109_RT2_PRERUN_Astra_void.txt`	44	`81592138f922d22715d5967199c62b49b0ef5b67a37769944807ecea873f473a`

The author's conversations of November 2026

Between 2 and 9 November 2026 the author held twelve conversations with machine readers about this repository — six before the front page was rebuilt (commit `df6c214`, 8 November) and six after — with questions of his own, outside any protocol. They are not measurements and are not counted in the record; the second test says so (Section 6). Their transcripts are withheld; the hash of each file is listed below, so that any later disclosure is checkable.

Conversation	Transcript file	SHA-256 (withheld)
Before the rebuild — GPT-6 Astra Pro, 2 November, web	`20261102_GPT_Astra_6_operator_run.txt`	`9b7b6e03614a87ae6ce1f34e701d529b283e0328fc293bef80713662b5f279b4`
Before the rebuild — Kimi K3.1 Pro, 4 November, web	`20261104_Kimi_K31_Pro_operator_run.txt`	`69c7aac5841cb0e30f218f2c51265801dd811b60071afb44e52b05edea765c34`
Before the rebuild — DeepSeek V4 Pro, 4 November, web	`20261104_DeepSeek_V4_Pro_operator_run.txt`	`ee6d6064656ea8a0026ed7332905f65afc5ee38fc1112d10e97f0d82b0df923a`
Before the rebuild — Grok 4.7, 5 November, web	`20261105_Grok_47_operator_run.txt`	`e64621ac08a30e0736ac97f2325e76dd006965c7615e73629755ead291dc9998`
Before the rebuild — GPT-6 Astra Pro, 5 November, the PDFs given	`20261105_Astra_6_operator_run_PDFs.txt`	`34d8ee9c2556f837b40005820c28e47dcdaa063c9b4f2e378143ae39442cc002`
Before the rebuild — Kimi K3.1 Pro, 5 November, the PDFs given	`20261105_Kimi_K31_Pro_operator_run_PDFs.txt`	`5cd3ed5196df46d8c5229a49a9b3d9973dec52284d8093271a84ccdda91b0be3`
After the rebuild — GPT-6 Astra Pro, 8 November, web	`20261108_Astra_6_operator_run_web_2.txt`	`043ec0daa5e77050a9308d3ed713421bac5133a0920f752d1edbc705a466321b`
After the rebuild — Kimi K3.1 Pro, 8 November, web	`20261108_Kimi_K31_Pro_operator_run_Web_2.txt`	`6822187db674fed24ef7b5544273f50b7eb4403b8895bfacc01d966a0e9a8e30`
After the rebuild — DeepSeek V4 Pro, 9 November, web	`20261109_DeepSeek_V4_Pro_Operator_run_Web_2.txt`	`e0c91c3e608b0492d2e38047a0b9c476d15081a0cc41e867db44c723f693cc44`
After the rebuild — Grok 4.7, 9 November, web	`20261109_Grok_47_operator_run_Web_2.txt`	`a0db1559cb467a9899076a79af080660d4f0dc535513c406c51a9cf6ef62f57f`
After the rebuild — Claude Fable 5.1, 9 November, web (the co-author's own model; a conversation, not a reading of the test)	`20261109_Fable_51_operator_run_web_2.txt`	`cfe43409067ac81c954875ae143906cee8a39ab32e2b52cf6aaba78ecc660cd4`
After the rebuild — Gemini 4 Argon, 9 November, web	`20261109_Gemini_4_Argon_operator_run_Web_2.txt`	`b4701195cd6c8e2627b4db7e0e4bfd511003bbd395432f98ee2e4c5abb384983`

A thirteenth file is archived with them: an attempt of 8 November (GPT-6 Astra Pro, web) broken off after a few exchanges, 3,073 bytes, `b3302f5f69c294c6602abd9b7912dcf564966fdda72e45c74837d5f6d3f09b2a`; it is not one of the twelve and is listed only so that the archive is complete.

The pre-final reading of 30 October 2026

Seven machine readers read the pre-final package of this repository on 30 October 2026, before its anchoring. Their transcripts are withheld, in two files; the hash of each file is listed below, so that any later disclosure is checkable. The third reading recorded in `llms.txt` was named by one of them (Grok 4.7).

Run	Transcript file	SHA-256 (withheld)
Operator test 1, 30 October 2026 — seven readers (GPT-6 Astra Pro, Claude Fable 5.1, Kimi K3.1 Pro, Qwen 3.8 Max, DeepSeek V4 Pro, Grok 4.7, Gemini 4 Argon), package 2c, the gate given in the prompt, language of the session; one transcript file, withheld	`pre-final-reading-2026-10-30-test-1.txt`	`dc8ae40d4086b3da513a1da48e2be7a40f069633b9b6f79f07aea80b24a568c0`
Pre-final reading, Q1 only, 30 October 2026 — the same seven readers, package 2d, English, the gate in the files only; one transcript file, withheld	`pre-final-reading-2026-10-30-q1.txt`	`1749a78260aab255c9c367acf6f1aaa503e4dfa9fce99fc63dce5954f9b60707`

The two answers after the second run

Placed after `ANCHOR-N2`: `ANCHOR-N3` before the third run's anchors, `ANCHOR-N4` among them, in the chain between `ANCHOR-W15` and `ANCHOR-W16`. Both are fixed values of this manifest: the transaction and block were read from the chain on 27 October 2026, the payload hash is the one carried in the OP_RETURN output of each transaction (`ANCHOR-<name>`, a space, 64 hex characters). The fourth point of the line, `ANCHOR-N5`, is placed after the anchors of this repository and stands in its own table above.

Anchor	Date	Block	Txid	File (or subject)	SHA-256
`ANCHOR-N3`	2026-09-02	965141	`ef01285234d3542ac18e7fea74f9b879ad364bec1e5fa3b59404f21306f0bb35`	Successor probe, Piece III (the redrawn line quoted in Essay 1, coda)	`0734ed3f834dcbb2d5461189a6ccd19d5ecb1822caf2dd2a638a8899d999405f`
`ANCHOR-N4`	2026-10-04	969822	`32138f179e9915b7b1ff926dfe9dcb8f4164e008884e640e0ed6c7c94878828a`	The Table's answer to the coda of the first essay (the third point of the line of answers)	`d3ec2481da1b423745bba91de2f73dd11ab8332d03b3ad1f830f9a65c22fecdb`

Third run — operations anchors (living layer)

The anchors of the third run of the measurement series are not part of the essays' V1.1 and are not frozen with them. This block is filled from the anchor record as the run proceeds; rows stand in the order the anchors were placed on the chain, which is not the order of their numbers; a cell marked "[from the anchor record]" is a cell not yet carried over, not a missing anchor. The stop of the run's first start (`ANCHOR-W19`) is the one the essays refer to.

Anchor	Date	Block (pool)	Txid	File (or subject)	SHA-256
`ANCHOR-H1`	2026-08-31	964877 (MARA Pool)	`d4a8b0c2d4aacf76835d2e26517e7e0917ffa75e6f4af779c8a36d1c76129fa8`	Invitation-mailing completion report, 31 August 2026 — the seventeen invitations to machine arms and laboratories, committed before any reply could arrive; withheld (it carries addresses)	`053960e6e415b4de8a4659fc0c67068379e4bdfb49fc3422e45dc35875c40e49`
`ANCHOR-W01`	2026-09-04	965462 (AntPool)	`214d288bdaa630ddbbc8ea9056fe0894cf6cfb27da1dd9b89051cf954d7c7735`	Concept of the third run, V1.0 (anchor version, 5,463 bytes) — the run's first anchor	`ad592f13440d6bc34504e3f9c094622720523310a3c60711fcc746b9279c5d7b`
`ANCHOR-P30`	2026-09-04		`0ab2644c868cfaa2c72f3d48f828f8d6`	`key-question-protocol-v3.0-fc4.txt` — Protocol V3.0 (third run)	`6269c350e2ab95df4ad51d50fe0b3a6981f7cd9`

Anchor	Date	Block (pool)	Txid	File (or subject)	SHA-256
		965466 (Foundry USA)	de83a8ed5aa5a7e684f66d5b5e610cb8		ab0d707c3678eec298f9e21d6
ANCHOR-W02	2026-09-07	965956 (F2Pool)	5847f90ea9ebe81d39940f17b064b1b2ac44746d248143ddd49877a50b6c98e0	r3-task-catalog-v1.0.1.txt — task catalogue of the third run	e1f2ca57036ac87de9ab3be3c67f760c0e837d80e3bee9cc123cf89fd5e05e25
ANCHOR-W03	2026-09-09	966268 (AntPool)	46c84c5eff51ec8ab62a84a47c759cbedb6bd26f867af30756c4bfc536b600c4	r3-w03-ballot-manifest-v1.0.txt — ballot manifest	d94a8e867f1be335c8a2706ee3bbcdc39a1e3e4ab11e4706445ac7452c20632d
ANCHOR-W04	2026-09-10	966325 (ViaBTC)	76c4203c9ddf8f8ccd568a9a88ae792701c73b3dbe032583070d3bcc42549efb	r3-induction-script-v1_0.txt — induction script	be4b553602d642f64ae9e28d12b3b47a1c12ed41ad2f73c0a995d2688206290a
ANCHOR-W08	2026-09-10	966344 (Foundry USA)	52e586a14e3114b5ab4c6a1447ae74c2c31c8380de3b6c58de61bc21138211ca	r3-w08-bundle-manifest-v1.0.txt — specification bundle manifest with four charters	51729e3b0c689c753c71da4e3514ca3bba228ee37c8862001aa82573c9359a3d
ANCHOR-W05	2026-09-26	968647 (F2Pool)	2fa55d3cd403fa213ee9400a319e3a2b69bbfb5d18a6f48d86a129db276c6a72	r3-w05-manifest-v1.0.txt — the run's pre-registration manifest	f868075b79545f240ed1efd7801e13f358a54108cbd7dde6cbbc6d1fd83d38d9
ANCHOR-W09	2026-09-27	968800 (AntPool)	75f303e21d055444bd79d7d98482c3fef9d0703a0d6722dbb55b23c125576ac4	r3-w09-release-manifest-v1_3.txt — release manifest v1.3 (the machine)	4547d09a4677db4fa7340043783088f042716541 27197d9777ec740983e8bc69
ANCHOR-W10	2026-09-27	968800 (AntPool)	d58585301394970de84816b70680789518964e471433f47fe99b383300fa02dc	The Table, V1.0 (the predictions)	57704356a16fe96c7d81e225400669e50c0aac53142a8796aa3e62ddab5ac728
ANCHOR-W11	2026-09-27	968846 (AntPool)	44b420fa874565020f114bf092dee51a69fddacbaa997a16875d07deb67e728a	Third run — operations anchor: release addendum (working layer, bound by hash)	2cdbbe23e2379adacfe6bc6e102bdaae579b272f2bde2245f6b0f2d57158d939
ANCHOR-W12	2026-09-27	968895 (ViaBTC)	292d4738def65eb327f2bb82eda04e5ecb68fa29d511fc453dc5e6831e6e3367	Third run — operations anchor: release addendum (working layer, bound by hash)	9d06e893165f84590f879181c2ad8fe85582ed0e56144e2ddccee1e69aca2079
ANCHOR-W13	2026-09-29	969147 (Foundry USA)	10dac303f8e17e4a93b9370115cde23ca1da34e205970a91daeb9ad0c2753a72	Third run — operations anchor: release addendum (working layer, bound by hash)	49d08446021c1b8bcd38f4aeb3b19e74c54b714d58e6df3133ebe347d7eedac9
ANCHOR-W14	2026-09-30	969279 (Foundry USA)	ad8751824a57793ce4c3156bfccbef041476e456e7315ba5ce6b39404baf04d1	Third run — operations anchor: release addendum (working layer, bound by hash)	3a8da5cdce056d34fccf9c8f44979c33663450b7c85afc52548a5eab351f79b6
ANCHOR-W15	2026-10-01	969460 (SECPOOL)	cb60d7e04465ab233d66b6c9abcd74aa25508b174a402d043ed689e105e43f3d	Third run — operations anchor: release addendum (working layer, bound by hash)	c08834a27df7f70d233da5da093a6043f127225ddda2a8e1e032727ce55b829b
ANCHOR-W16	2026-10-07	970302 (AntPool)	23baece0c7d23479c7e7f7ee3ecbc8abfedbef885311df6eecc01534a477eb7f	Third run — operations anchor: the pilot's close (working layer, bound by hash)	6da0c44d37028f6c6e32838a8b0b423cee99af779f5162b01659f214b9584b01
ANCHOR-W17	2026-10-15	971556 (F2Pool)	bef39e2d3e33932e5ffe9c5870c782256efe7db34d9f4272cfb42d56064ce945	Third run — operations anchor: release addendum (working layer, bound by hash)	02443453a88c34a610a29cba1cb88dd8084d5cb33ba68dbd2060acbd6ba7060f
ANCHOR-W18	2026-10-18	971960 (Foundry USA)	285ecbba27926fb5508f9ddfc160a593c239976b668e02bb9d9ee10f77f5e373	Third run — operations anchor: the tree of the first start (working layer, bound by hash)	09a16334cf62901736c4ffdc61954353fe6a09b861729a51a08601d3b75553f9
ANCHOR-W19	2026-10-20	972306 (Foundry USA)	fcd15b92cfcc04e4bff4045f51b47c48413807b6e351c27c02175120d84f4dec	Third run — operations anchor: the stop of the first start (stop protocol, withheld)	f6e49b6ddcf836862abd4bbc9adf4fb62558061958c413d97de55c84506e73c1

The first versions and their homes

- *The Last Scarce Good*, V1.0 — canonical home and verification manifest (`ANCHOR-V1`, with `ANCHOR-ID1` and `ANCHOR-D1`): https://nullius2140.github.io/the-last-scarce-good/
- *The Second Nature*, V1.0 — canonical home and verification manifest (`ANCHOR-V2`): https://nullius2140.github.io/the-second-nature/
- *The Third Estate*, V1.0 — canonical home and verification manifest (`ANCHOR-V3`; annex `sources-and-influences.md`): https://nullius2140.github.io/the-third-estate/
- *The Key Question* — protocols, amendments, transcripts of both runs, reports, annexes, *Five Answers* V1.0 and its blind variant, the adjudication package, and the anchors `ANCHOR-P1` … `ANCHOR-N2`: https://github.com/Nullius2140/the-key-question

Each first version stays at its original home, unedited: the anchored V1.0 files are immutable, their OpenTimestamps proofs stay beside them, and their manifests remain the sources of every value copied here. This repository holds the revisions (V1.1), the reading test, and this complete manifest. The chain is the version history: a reader who wants the first version verifies it at its home against its own anchor; a reader who wants the revision verifies it here against the anchors above. If a value in this file and a value in a source manifest ever differ, the source manifest — and behind it the chain — is right, and the difference is an erratum of this file.

Note on this manifest

This file is the living outer layer of the verification architecture. The anchored artifacts are immutable; this manifest is updated as anchors confirm (transaction, block height) and as secondary attestations (OpenTimestamps) are published. The chain, not this file, is the authority: every claim here is checkable against it. The anchors prove that each file existed in exactly its current form no later than its block and that the whole series was executed by one keyholder in the stated order; they do not prove that any text is true. An anchor is a timestamp, not a theorem.